hw3

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This is the link to the repo: https://github.com/YifengHuang-Bob/Homework3

Problem 1

(a) Download and merge the datasets

```
library(haven)

aux <- read_xpt("data/AUX_I.xpt")
demo <- read_xpt("data/DEMO_I.xpt")

nhanes <- merge(aux, demo, by = "SEQN")

dim(nhanes)
```

```
[1] 4582  119
```

(b) Data cleaning

```
nhanes$RIAGENDR <- factor(
  nhanes$RIAGENDR,
  levels = c(1, 2),
  labels = c("Male", "Female")
)
```

```
nhanes$DMDCITZN[nhanes$DMDCITZN %in% c(7, 9)] <- NA

nhanes$DMDCITZN <- factor(
  nhanes$DMDCITZN,
  levels = c(1, 2),
  labels = c("Citizen", "Not a citizen")
)
```

```r
nhanes$INDHHIN2[nhanes$INDHHIN2 %in% c(77, 99)] <- NA

nhanes$INDHHIN2 <- factor(
  nhanes$INDHHIN2,
  levels = c(1:10, 14, 15, 13, 12),
  labels = c(
    "$0-$4,999",
    "$5,000-$9,999",
    "$10,000-$14,999",
    "$15,000-$19,999",
    "$20,000-$24,999",
    "$25,000-$34,999",
    "$35,000-$44,999",
    "$45,000-$54,999",
    "$55,000-$64,999",
    "$65,000-$74,999",
    "$75,000-$99,999",
    "$100,000 and over",
    "Under $20,000 (broad)",
    "$20,000 and over (broad)"
  )
)
```

The original income codes do not follow a strictly increasing income order. I rearranged the detailed income categories from lowest to highest and placed the two broad, overlapping categories separately at the end, as they cannot be precisely ordered among the detailed income categories.

(c) Poisson regression models

```r
nhanes$income_numeric <- as.numeric(nhanes$INDHHIN2)

nhanes$income_numeric[nhanes$income_numeric > 12] <- NA
```

```r
model_1r <- glm(
  AUXTWIDR ~ RIAGENDR,
  data = nhanes,
  family = poisson(link = "log")
)
```

```r
model_2r <- glm(
  AUXTWIDR ~ RIAGENDR + DMDCITZN + DMDHHSZA + income_numeric,
```

```
  data = nhanes,
  family = poisson(link = "log")
)
```

```
model_1l <- glm(
  AUXTWIDL ~ RIAGENDR,
  data = nhanes,
  family = poisson(link = "log")
)
```

```
model_2l <- glm(
  AUXTWIDL ~ RIAGENDR + DMDCITZN + DMDHHSZA + income_numeric,
  data = nhanes,
  family = poisson(link = "log")
)
```

```
irr_table <- data.frame(
  Predictor = c(
    "Intercept (exp(coefficient))",
    "Female",
    "Not a citizen",
    "Children (per unit)",
    "Income level (per unit)"
  ),
  `1R` = c(exp(coef(model_1r)), rep(NA_real_, 3)),
  `2R` = exp(coef(model_2r)),
  `1L` = c(exp(coef(model_1l)), rep(NA_real_, 3)),
  `2L` = exp(coef(model_2l)),
  check.names = FALSE
)

options(knitr.kable.NA = "-")
```

```
knitr::kable(
  irr_table,
  row.names = FALSE,
  digits = 4,
  caption = "Exponentiated coefficients from the four Poisson regression models"
)
```

Predictor	1R	2R	1L	2L
Intercept (exp(coefficient))	84.3174	88.4374	84.7584	88.8213
Female	1.0093	1.0150	1.0130	1.0187
Not a citizen	-	1.0469	-	1.0218
Children (per unit)	-	0.9944	-	0.9830
Income level (per unit)	-	0.9921	-	0.9935

```
model_stats <- data.frame(
  Statistic = c("Sample size", "Pseudo-R2 (deviance-based)", "AIC"),
  `1R` = c(
    format(nobs(model_1r), big.mark = ","),
    sprintf("%.6f", 1 - deviance(model_1r) / model_1r$null.deviance),
    format(round(AIC(model_1r)), big.mark = ",")
  ),
  `2R` = c(
    format(nobs(model_2r), big.mark = ","),
    sprintf("%.6f", 1 - deviance(model_2r) / model_2r$null.deviance),
    format(round(AIC(model_2r)), big.mark = ",")
  ),
  `1L` = c(
    format(nobs(model_1l), big.mark = ","),
    sprintf("%.6f", 1 - deviance(model_1l) / model_1l$null.deviance),
    format(round(AIC(model_1l)), big.mark = ",")
  ),
  `2L` = c(
    format(nobs(model_2l), big.mark = ","),
    sprintf("%.6f", 1 - deviance(model_2l) / model_2l$null.deviance),
    format(round(AIC(model_2l)), big.mark = ",")
  ),
  check.names = FALSE
)

knitr::kable(
  model_stats,
  row.names = FALSE,
  caption = "Model statistics for the four Poisson regression models"
)
```

Statistic	1R	2R	1L	2L
Sample size	4,149	3,705	4,103	3,665
Pseudo-R2 (deviance-based)	0.000106	0.005717	0.000200	0.003631
AIC	96,618	84,916	98,685	86,598

(d) Hypothesis tests

```
gender_test <- coef(summary(model_2l))["RIAGENDRFemale", ]

gender_test
```

```
    Estimate   Std. Error      z value     Pr(>|z|)
1.857273e-02 3.599573e-03 5.159705e+00 2.473393e-07
```

```
exp(coef(model_2l)["RIAGENDRFemale"])
```

```
RIAGENDRFemale
      1.018746
```

```
emm_gender <- emmeans::emmeans(
  model_2l,
  ~ RIAGENDR
)

summary(emm_gender, type = "response")
```

```
 RIAGENDR rate    SE  df asymp.LCL asymp.UCL
 Male     84.8 0.252 Inf      84.3      85.3
 Female   86.3 0.249 Inf      85.9      86.8

Results are averaged over the levels of: DMDCITZN
Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

```
gender_diff <- pairs(
  emmeans::regrid(emm_gender, transform = "response"),
  reverse = TRUE
```

```
)

summary(gender_diff, infer = c(TRUE, TRUE))
```

```
 contrast      estimate    SE  df asymp.LCL asymp.UCL z.ratio p.value
 Female - Male     1.59 0.308 Inf     0.986      2.19   5.161 <0.0001

Results are averaged over the levels of: DMDCITZN
Confidence level used: 0.95
```

Test 1 — IRR Comparison

For Model 2L, the estimated female-to-male IRR is 1.0187. The Wald test gives $z = 5.160$ and $p = 2.47 \times 10^{-7}$. Therefore, we reject the null hypothesis that the IRR equals 1 at the 5% significance level. Holding other covariates constant, females have approximately 1.87% higher expected left-ear tympanometric width than males.

Test 2 — Predicted Value Comparison

The adjusted predicted values are 84.8 for males and 86.3 for females. The estimated difference (Female – Male) is 1.59, with a 95% confidence interval of $[0.986, 2.19]$. The test gives $z = 5.161$ and $p < 0.0001$. Since the confidence interval excludes 0, we reject the null hypothesis of equal predicted values at the 5% significance level.

Note on Overdispersion

These results are based on the standard Poisson model. Since the model shows evidence of overdispersion, the reported p-values and confidence intervals should be interpreted cautiously.